Please check that this question paper contains____09__ questions and _2_____ printed pages within first ten

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07 DEC 2023

[Total No. of Questions: 09]

Uni. Roll No. ……………

[Total No. of Pages: ...02...]

Program: B.Tech. (Batch 2018 onward)

Semester: 3rd

Name of Subject: Digital Circuits and Logic Design

Subject Code: ESIT-101

Paper ID: 16042

Time Allowed: 03 Hours

Max. Marks: 60

NOTE:

1) Parts A and B are compulsory
2) Part-C has Two Questions Q8 and Q9. Both are compulsory, but with internal choice
3) Any missing data may be assumed appropriately

Part – A

[Marks: 02 each]

Q1.

a. Convert the binary number 11010101 into its corresponding hexadecimal form.

b. State De- Morgan's Theorem.

c. What are the various uses of VHDL?

d. How multiplexer different from decoder?

e. Differentiate Moore and Mealy circuits.

f. Define race condition in JK flip flop

Part – B

[Marks: 04 each]

Q2. Define 2's complement. Subtract $(1010)_2$ from $(1000)_2$ using 2's complement method.

Q3. Write a note on RTL logic family.

Q4. Illustrate the working of Master Slave J-K flip flop.

Q5. Explain the full adder circuit using logic diagram and Truth Table.

Q6. Describe the working of Johnson Ring Counter.

Q7. Illustrate the operation of basic flip-flop using NOR gates.

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Part – C [Marks: 12 each]

Q8. Write short note on following:

a) Successive approximation A to D conversion technique (6 marks)

b) Ripple Carry Adder (6 marks)

OR

Explain different logic gates families in digital circuits. Write a short note on Universal Gate.

Q9. Design 2-bit Synchronous Up counter using JK flip flop.

OR

Minimise the following function in SOP minimal form using K-Maps: F (A, B, C, D) = m (1, 2, 6, 7, 8, 13, 14, 15) + d (0, 3, 5, 12)

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